DEPARTMENT OF SOFTWARE ENGINEERING
BACHELORS IN SOFTWARE ENGINEERING

Complex Engineering Problem

Course Code: SE-204, Course Title: Database Management System

CLO: CLO 2 and CLO 3

Course Learning Outcome

CLO 2: Explore database development and administration problems (Taxonomy level: C3), PLO 2(Problem Analysis)

CLO 3: Prepare students to works as a valuable member of a database design and implementation team (Taxonomy level C3), PLO9 (Individual and Team Work)

Complex problem solving attributes covered (as per PEC - OBA manual – 2019)

- Range of resources: Involve the use of diverse resources (and for this purpose, resources include people, money, equipment, materials, information and technologies).
- Level of interaction: Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering or other issues.
- Innovation: Involve creative use of engineering principles and research-based knowledge in novel

Problem Statement

Apply database design and development concepts to develop a database management system (DBMS) solution.

Description

You will select a project from those provided, analyze the organizational issues and needs, and work your way through the conceptual, logical, and physical designs of a DBMS solution. This will require substantial research of best practices in design, available products, and the legal and ethical standards to which you must adhere during design. The skills required in this assessment will be valuable in the role of an IT manager or DBMS professional, as these individuals are often tasked with developing solutions to various organization data problems while also adhering to legal, ethical, and financial considerations. The ability to craft and present professional and logical proposals will also be expected of IT professionals, especially those at the management level.

The following critical elements must be addressed:

I. Organization

A. Problem: Analyze the organization to determine the problem/challenge, business requirements, and limitations of current system(s).

B. Departments and Operations: Explain how the various departments and operations within the organization are impacted by the issue or challenge

II. Analysis and Design

A. Conceptual Model: Based on the business problem or challenge, devise a conceptual model that would best address the problem. Your model should include all necessary entities, relationships, attributes, and business rules.

B. Logical Model: Based on the conceptual model, illustrate a logical model for your DBMS that accurately represents all necessary aspects of the DBMS to address the solution.

C. Physical Design: Create a physical database design that builds on the nonphysical (conceptual and logical) models you crafted.

Design Constraints

- The database will be designed to have multiple records and fields.
- At least two different field types from the following list must be included: numeric, alpha, date/time, and currency.

Deliverables

- Prepare a report and bring it at the day of your viva. The report has to be organized as follows:
    - o Title page. The rubric sheet given at the end of this document will serve as the title page of your report.
    - o Text of 500 words at maximum, containing the following:
        1. Business requirements, problem/challenge, and limitations of current system
        2. Participation in a team
        3. ER and EER Diagram of your project.
        4. Functional Dependencies
        5. Process of Normalization (1NF to BCNF)

DEPARTMENT OF SOFTWARE ENGINEERING
BACHELORS IN SOFTWARE ENGINEERING
Course Code: SE-204
Course Title: Database Management System
Complex Engineering Activity
SE Batch 2024, Spring Semester 2026
Grading Rubric

Group Members:

Student No.	Name	Roll No.
S1		
S2		
S3		

CRITERIA AND SCALES			Marks Obtained		
			S1	S2	S3
Criterion 1: Does the problem statement/ motivation clearly stated?					
0 – 1	2 - 3	4 – 5			
Objective of the project are not clear or described.	Some lack of clarity in objectives/purpose.	The project's objectives are clearly stated.			
Criterion 2: How well is the Entity Relationship Model?					
0 – 1	2 - 3	4 – 5			
The model does not attend to all necessary entities, relationships, attributes, and business rules or would not logically address the identified problem	The model is a comprehensive conceptual model that would logically address the identified problem	The model Meets "Proficient" criteria, and model evidences detailed attention or keen insight into conceptual design needs and skills			
Criterion 3: How well is the Enhanced Entity Relationship Model?					
0 – 1	2 - 3	4 – 5			
The model does not attend to all necessary sub classes and super classes, specialization and generalization and aggregation concepts	The model is a comprehensive conceptual model that would logically address all necessary sub classes and super classes, specialization and generalization and aggregation concepts.	The model meets "Proficient" criteria, and model evidences all necessary sub classes and super classes, specialization and generalization and aggregation concepts.			
Criterion 4: How well the relations are Normalized?					
0 – 1	2 - 3	4 – 5			
Relations are not normalized to Boycee Codd form.	Relations are normalized to Boycee Codd normal form	Relations are correctly normalized to Boycee Codd normal form.			
Criterion 5: How did the student answer questions relevant to the task?					
0 – 1	2 - 3	4 – 5			

Student was not able to apply foreign keys accurately, did not follow relationship protocol.	Student adequately demonstrates understanding of foreign keys and primary keys, but does not follow relationship protocol with one of the Directories.	Student adequately demonstrates understanding of foreign keys, primary keys, and relationship protocol by creating a relationship table with foreign keys identified.			
Criterion 6: How did the team works?					
0 – 1	2 - 3	4 – 5			
Team did not collaborate or communicate well. Some members would work independently, without regard to objectives or priorities. A lack of respect and regard was frequently noted.	The team worked well together most of the time, with only a few occurrences of communication breakdown or failure to collaborate when appropriate. Members were mostly respectful of each other.	The team worked well together to achieve objectives. Each member contributed in a valuable way to the project. All data sources indicated a high level of mutual respect and collaboration			
Criterion 7: How did the student participate individually in a team?					
0 – 1	2 - 3	4 – 5			
The individual did not contribute to the project and failed to meet responsibilities. The individual does not identify key performance criteria of successful teams or draw inference to own experience	The individual did not contribute as heavily as others but did meet all responsibilities. The individual is also able to identify some key performance criteria of successful teams and/or draw related connections the group performance	The individual contributed in a valuable way to the project. The individual is also able to articulate the key performance criteria of successful teams and evaluate the group performance accordingly			
		Total Marks:			

Teacher's Signature:____________